Project Title

First Name Last Name

Project Title

Thesis submitted to in partial fulfillment
of the requirements for the degree

of

Bachelor of Technology

in

Electronics & Communication Engineering (ECE) (or CSE)

by

First Name Last Name (Student)

Under the guidance of

Dr. First Name Last Name

Designation

Name of the Institute

Dr. SPM International Institute of Information Technology
Naya Raipur, India 493661
Month year (of submission)

Your Dedication
(Ex: This work is dedicated to my parents, Family and Friends for their Support, Sacrifice Encouragement and love)

Approval of the Viva-Voce Board

April 29, 2019

Certified that the thesis entitled **Thesis Title** submitted by **Name of the student** to the Dr. SPM International Institute of Information Technology, Naya Raipur, India, for the award of the degree of Bachelor of technology has been accepted by the examiners and that the student has successfully defended the thesis in the viva-voce examination held today.

(Name)
Designation, IIIT-Naya Raipur.

(Name)
Designation, IIIT-Naya Raipur.

(Name)
Designation, IIIT-Naya Raipur.

(Name)
Designation, IIIT-Naya Raipur.

(Name)
Designation, IIIT-Naya Raipur.

DECLARATION

April 29, 2019

I certify that

a. The work contained in the thesis is original and has been done by myself under the general supervision of my supervisor.

b. The work has not been submitted to any other Institute for any degree or diploma.

c. I have followed the guidelines provided by the Institute in writing the thesis.

d. I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

e. Whenever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references.

f. Whenever I have quoted written materials from other sources, I have put them under quotation marks and given due credit to the sources by citing them and giving required details in the references.

(Name of the student with signature)

Date: April 29, 2019

Certificate

This is to certify that the thesis entitled, **Thesis Title** submitted by **Name of the student** to Dr. SPM International Institute of Information Technology, Naya Raipur, Chhattisgarh, India, is a record of bona fide research work under my supervision and I consider it worthy of consideration for the award of the degree of Bachelor of Technology of the Institute.

Signature of the Supervisor: ....

Name of the Supervisor: **Dr. First Name Last Name**,

Designation:
Institute:

Acknowledgment

Acknowledgment not more than two pages.

April 29, 2019
Naya Raipur

(Name of the student with signature)

Plagiarism Report

Attache plagiarism Report not more than two pages.

Abstract

Aabstract of the Major project thesis not more than two pages.

Keywords: Maximum five numbers of Key wordes according to alphabet.

Contents

List of Abbreviations

ABC	:	absorbing boundary condition
AF	:	antenna factor
BC	:	boundary condition
CAF	:	complex antenna factor
CSD	:	corner slope diffraction
EDSB	:	edge-diffraction shadow boundary

List of Notations and Operations

List of symbols

B : magnetic flux density vector

D : electric flux density vector

J_c : conduction current density vector of a lossy medium

ε : electric permittivity of medium

ε_0 : electric permittivity of free space ($8.854 \times 10^{-12}\ F/m$)

$\frac{d^2}{dt^2}$:second derivative with respect to t

$\bigtriangledown^2$:Laplace operator

List of Figures

List of Tables

CHAPTER 1

Introduction

Preface

This chapter presents the introduction of the thesis and a survey of the literature on the problems. The gaps in the works reported so far, as well as the unexplored areas are also dealt with. The methodology to compute is also mentioned here. How the thesis is organized is also detailed in this chapter.

1.1 Introduction

Introduction of your whole thesis. The introduction to your thesis is the first thing the examiner will read. Its your only chance to form a first impression, if the examiner doesn't already know you. It sets the background, context and motivation for your work. And so its at least as important as every other chapter.

1.2 Review of Literature

If you have to write an undergraduate dissertation, you may be required to begin by writing a literature review. A literature review is a search and evaluation of the available literature in your given subject or chosen topic area. It documents the state of the art with respect to the subject or topic you are writing about.

A literature review has four main objectives:

(i) It surveys the literature in your chosen area of study.

(ii) It synthesises the information in that literature into a summary.

(iii) It critically analyses the information gathered by identifying gaps in current knowledge; by showing limitations of theories and points of view; and by formulating areas for further research and reviewing areas of controversy.

(iv) It presents the literature in an organised way.

A literature review shows your readers that you have an in-depth grasp of your subject; and that you understand where your own research fits into and adds to an existing body of agreed knowledge.

1.3 Objectives and Scope

Following the survey of various available published literatures we propose

(i) Objectives and (/or) scope-1.

(ii) Objectives and (/or) scope-2.

(iii) Objectives and (/or) scope-3.

(iv) Objectives and (/or) scope-4.

1.4 Methodology Followed

This article is about your research methods.

Methodology is the systematic, theoretical analysis of the methods applied to a field of study. It comprises the theoretical analysis of the body of methods and principles associated with a branch of knowledge. Typically, it encompasses concepts such as paradigm, theoretical model, phases and quantitative or qualitative techniques.

It has been defined also as follows:

(i) the analysis of the principles of methods, rules, and postulates employed by a discipline.

(ii) the systematic study of methods that are, can be, or have been applied within a discipline.

(iii) the study or description of methods.

(For example: For GTD computation programs are developed in $FORTRAN - 90$ using $gcc - 4.0$ compiler. Whereas for the numerical calculation, a program based on FDTD technique is developed in C using compiler $gcc - 4.0$ run on a Pentium 2.3 GHz processor based on personal computer supported by $LINUX$ operating system.)

1.5 Organization of the Thesis

This dissertation shows how a realistic model that can help others to understand your research work. The organization of this thesis is as follows.

Chapter 1 deals with the introduction of the thesis. It contains a brief review of the earlier works on the problem and the present approach using. It provides a introduction to the problem. It also contains the objectives and scope of the work reported in the thesis and lastly, the contents of the thesis.

Chapter 2 contains an introduction to the your technique.....

Chapter 3 is devoted to the problem of.....

Chapter 4 deals with .....

Chapter 5 deals with the further analysis of ....

Chapter 6 deals with the .....

Chapter 7 deals with the discussions on all aspects of the work reported in this thesis. It also contains important conclusions & scope for future work.

CHAPTER 2

Equations and Figures

Preface

Preface of this chapter.

2.1 Introduction

Introduction of this chapter.

2.2 Examples of Equations

The differential form of Maxwell's equations and constitutive relations [1] can be written as:
Faraday's law:

$$\bigtriangledown \times \overrightarrow{E} = -\frac{\partial \overrightarrow{B}}{\partial t} - \sigma_m \overrightarrow{H} - \overrightarrow{M} \tag{2.1}$$

Ampere's Law:

$$\bigtriangledown \times \overrightarrow{H} = \frac{\partial \overrightarrow{D}}{\partial t} + \sigma_e \overrightarrow{E} + \overrightarrow{J} \tag{2.2}$$

Gauss's Law for the electric field:

$$\bigtriangledown \cdot \overrightarrow{D} = \rho_e \tag{2.3}$$

Gauss's Law for the magnetic field:

$$\bigtriangledown \cdot \overrightarrow{B} = \rho_m \tag{2.4}$$

Constitutive relations:

$$\overrightarrow{B} = \mu \overrightarrow{H} \tag{2.5a}$$
$$\overrightarrow{D} = \varepsilon \overrightarrow{E} \tag{2.5b}$$

In case of electromagnetic phenomenon in free space, or more general, in a perfect dielectric containing no charge and no conduction current, the field equations [2] become

$$\bigtriangledown \times \overrightarrow{H} = \frac{\partial \overrightarrow{D}}{\partial t} \tag{2.6a}$$
$$\bigtriangledown \times \overrightarrow{E} = -\frac{\partial \overrightarrow{B}}{\partial t} \tag{2.6b}$$
$$\bigtriangledown \cdot \overrightarrow{D} = 0 \tag{2.6c}$$
$$\bigtriangledown \cdot \overrightarrow{B} = 0 \tag{2.6d}$$

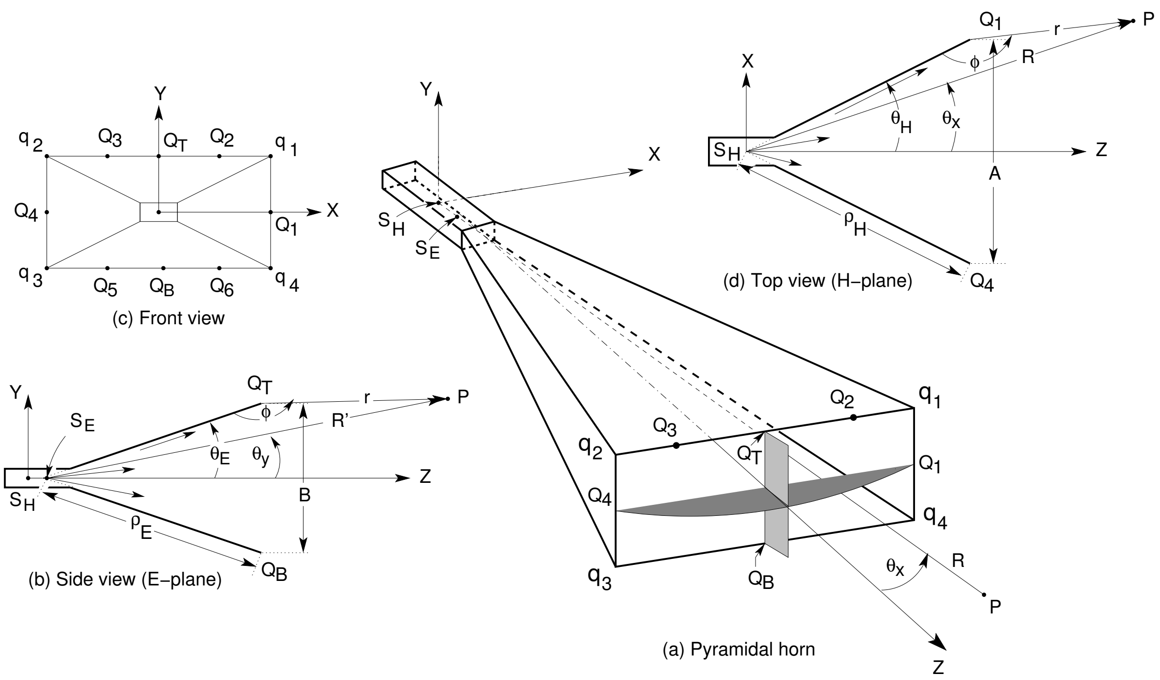

Figure 2.1: Geometry of a pyramidal horn used for GTD analysis.

Substituting (2.5a) and (2.5b) into (2.6a) and (2.6a), the Maxwell's curl equations become

$$\mu \frac{\partial \overrightarrow{H}}{\partial t} = - \bigtriangledown \times \overrightarrow{E} \tag{2.7}$$

$$\varepsilon \frac{\partial \overrightarrow{E}}{\partial t} = \bigtriangledown \times \overrightarrow{H} \tag{2.8}$$

In rectangular coordinates, (2.7) and (2.8) can be written as

$$\mu \frac{\partial H_x}{\partial t} = \frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} \tag{2.9a}$$
$$\mu \frac{\partial H_y}{\partial t} = \frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} \tag{2.9b}$$
$$\mu \frac{\partial H_z}{\partial t} = \frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \tag{2.9c}$$

2.3 Figures

Figures must appear in the text as near as possible to the discussion relating to them. Under no circumstances will a figure precede the first discussion of its content. (Exception: figures in Appendixes). Figs. 2.1 and 2.2 are the examples.

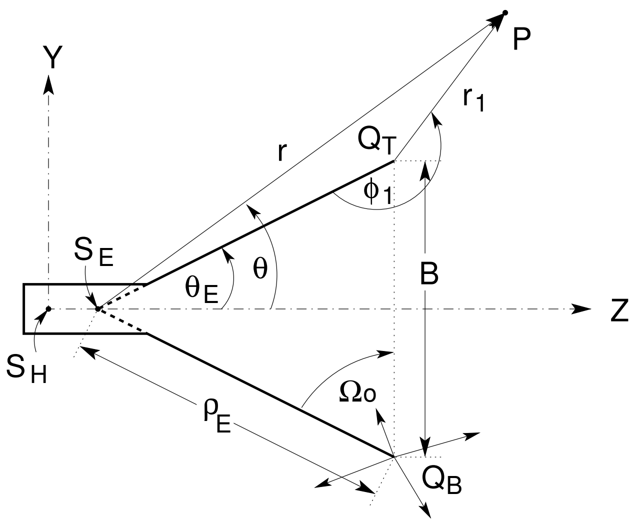

Figure 2.2: Geometry of the horn

2.4 References

Use IEEE referencing format. The IEEE style is a numeric style, where citations are numbered [1] in the order of appearance. This citation leads your reader to a full reference to the source in the list of references at the end of your thesis. Once a source has been cited, the same number is re-used for all subsequent citations to the same source. References must be numbered sequentially, not alphabetically.

2.5 Conclusions

Conclusions of this chapter.

CHAPTER 3

Conclusions & Scope for Future Work

Preface

The final chapter of this thesis sums up the achievements in this thesis, based on the analysis and results presented. It also critically examines the examines the areas which are not studied in the present work, and outlines the areas where efforts need to be devoted to further the state of knowledge in the area.

3.1 Conclusions

Conclusions of your whole thesis.

3.2 Contributions Made by the Scholar

(i) Contribution 1

(ii) Contribution 2

(iii) Contribution 3

3.3 Scope for Future Work

The future scope and possible extensions of the investigations carried out in this thesis are

(i) **Head line of scope for future work-1:**
Details of scope for future work-1

(ii) **Head line of scope for future work-2:**
Details of scope for future work-2

(iii) **Head line of scope for future work-3:**
Details of scope for future work-3

APPENDIX A

Thesis Format

Preface

Appendix A presents the thesis format guidelines.

The line spacing is multiple with 1.15 pt, with the indent at the new paragraph. Thesis should be printed in double side.

A.0.1 Thesis features

A new section paragraph starts with no indentation. However, there will be a regular indentation of 0.2"at the start of next paragraph.

The new paragraph can start in the next line, with no gap. There is an indentation of 0.2 at the beginning of the paragraph. Paragraph in the thesis will always be justified, not left-centered.

A.0.2 Body of the thesis

The thesis body should follow the following format.
Text font style: Times new Roman
Every chapter should be started with new odd page
Paper size: A4

A.0.3 Font

Chapter heading 20 pt bold
Heading 14 pt bold
sub heading 12 pt with bold
Text 12 pt
Line spacing: 1.15pt
Margin

(i) Top -1.5 "

(ii) Bottom- 1"

(iii) Left -1.5"

(iv) Right -1"

Header:

(i) Odd page - Thesis title

(ii) Even page - Chapter title

APPENDIX B

Instructions for Thesis Table

Preface

Appendix B presents for the format of table within thesis.

Tables must appear in the text as near as possible to the discussion relating to them. Under no circumstances will a table precede the first discussion of its content.

The numerical values of $a_n,\ b_n,\ c_n,\ d_n$ [3] are given in the Table B.1.

a_n	b_n	c_n	d_n
$a_0 = +1.595769140$	$b_0 = -0.000000033$	$c_0 = +0.0$	$d_0 = +0.199471140$
$a_1 = -0.000001702$	$b_1 = +4.255387524$	$c_1 = -0.024933975$	$d_1 = +0.000000023$
$a_2 = -6.808568854$	$b_2 = -0.000092810$	$c_2 = +0.000003936$	$d_2 = -0.009351341$
$a_3 = -0.000576361$	$b_3 = -7.780020400$	$c_3 = +0.005770956$	$d_3 = +0.000023006$
$a_4 = +6.920691902$	$b_4 = -0.009520895$	$c_4 = +0.000689892$	$d_4 = +0.004851466$
$a_5 = -0.016898675$	$b_5 = +5.075161298$	$c_5 = -0.009497136$	$d_5 = +0.001903213$
$a_6 = -3.050485660$	$b_6 = -0.138341947$	$c_6 = +0.011948809$	$d_6 = -0.017122914$
$a_7 = -0.075752419$	$b_7 = -1.363729124$	$c_7 = -0.006748873$	$d_7 = +0.029064067$
$a_8 = +0.850663781$	$b_8 = -0.403349276$	$c_8 = +0.000246420$	$d_8 = -0.027928955$
$a_9 = -0.025639041$	$b_9 = +0.702222016$	$c_9 = +0.002102967$	$d_9 = +0.016497308$
$a_{10} = -0.150230960$	$b_{10} = -0.216195929$	$c_{10} = -0.001217930$	$d_{10} = -0.005598515$
$a_{11} = +0.034404779$	$b_{11} = +0.019547013$	$c_{11} = +0.000233939$	$d_{11} = +0.000838386$

Table B.1: The numerical values of the coefficients of the Fresnel integral [3]

References

[1] Allen Taflove and Susan C. Hagness, *Computational Electromagnetic-The Finite-Difference Time-Domain Method*, 2nd ed. Boston London: Artech House, 2000.

[2] Edward C. Jordan, *Electromagnetic wave and radiating systems*, 2nd ed. New Delhi: Prentice-Hall of India Private Limited, 1995.

[3] J. Boersma, "Computation of fresnel integrals," *Math. Computation*, vol. 14, p. 380, 1960.

Index

Publications

Referred Journals:

- **Author 1** and Author 2, "..........", Scheduled to appear in *IEEE Transactions on Antennas and Propagation* 2019.

Conference Proceedings:

- **Author 1** and Author 2, "....................", in *Proc. of Applied Electromagnetic Conference, AEMC 2009, Hyatt Regency, Kolkata, India*, 14-16, April-2009.

Author's Biography

Name (Student) was born in ....., India on 21$^{\text{th}}$ December, 1999. ......

His current area of research includes .... He is associated with live projects .... During his research, he presented papers in many National and International Conferences and participated in important workshops. He can be contacted at: .....@lycos.com & ......@iiitnr.edu.in.